
PySpark
Interview
Questions

Pyspark Interview Questions

1. Spark Architecture & Core Concepts

1. Describe the architecture of Apache Spark and its components in a distributed environment.
2. What is PySpark, and how does it differ from Apache Spark?
3. Describe the difference between RDDs, DataFrames, and Datasets.
4. Explain the differences between transformations and actions in PySpark DataFrames with examples.
5. Explain the role of the SparkContext and how it differs from the SparkSession.
6. Explain lazy evaluation in PySpark and why it is important.
7. Can you describe how PySpark's DAG scheduler works and how it manages task execution?
8. What is DAG (Directed Acyclic Graph) and how does it help in Spark execution?
9. What is Spark and why is it preferred over traditional frameworks like MapReduce?
10. What is the role of Spark SQL in data processing and analysis?
11. What are the different deployment modes available in Spark (local, standalone, YARN, Mesos, Kubernetes)?
12. What is the difference between client mode and cluster mode in Apache Spark?
13. Explain the concept of lineage in Apache Spark and its significance in fault tolerance.
14. How does Apache Spark handle fault tolerance compared to other distributed computing frameworks?

2. Performance Optimization & Best Practices

1. How do you optimize the performance of PySpark jobs and what tuning techniques do you follow?
2. How does partitioning impact performance and how do you choose the correct number of partitions?
3. Explain broadcast variables and their role in PySpark optimization.
4. What are broadcast variables and accumulators in PySpark and how do they differ?
5. How do you handle skewed data in PySpark and what strategies do you use?
6. What is partition skew, what causes it, and how can it be mitigated?
7. Discuss techniques for handling skewed data in join operations.
8. What is the significance of caching in Spark and when should you use it?
9. What is the difference between persist() and cache() in PySpark?
10. What are the best practices for writing efficient and optimized PySpark code?
11. What is shuffling in Spark and why should we aim to minimize it?

12. How does Apache Spark handle memory management and garbage collection?
13. Discuss the significance of partitioning in Spark and how it affects performance.
14. What is the difference between partitioning and bucketing in Spark?
15. What are the optimizations performed by the Catalyst optimizer in Spark SQL?
16. Explain the Catalyst optimizer and its benefits for query performance.
17. Why is count used with groupBy a transformation, but otherwise an action?
18. How do you estimate the amount of resources required for your Spark job?
19. What are the various ways to persist data in Apache Spark?
20. Discuss the significance of choosing the right compression codec for your PySpark applications.

3. Real-World Scenarios & Debugging

1. A Spark job is running slower than expected - how do you debug and resolve it?
2. Identify and address performance bottlenecks in a slow-running Spark job.
3. If 199 out of 200 partitions are executed and 1 fails, what steps do you take to resolve it?
4. Handling a large dataset that doesn't fit into memory - how would you approach this scenario?
5. Joining two large datasets where one exceeds memory - how to optimize the join strategy?
6. If a job fails with an out-of-memory error in production, what will be your approach to debug and fix it?
7. What steps would you take to debug memory issues in PySpark?
8. How do you handle iterative computations in Spark such as ML and graph algorithms?
9. How would you process customer reviews to find the top N products with the highest average ratings?
10. How do you test your Spark code and ensure it behaves as expected?
11. How do you handle your PySpark code deployment and explain the CI/CD process?
12. What are some common errors you encountered in PySpark and how did you resolve them?
13. What methods or tools do you use for testing PySpark code effectively?
14. Have you used caching in your project? When and where do you consider using it?

4. DataFrame & RDD Operations

1. How do you create a DataFrame in PySpark?
2. What are transformations and actions in PySpark and how do they differ?
3. How do you perform joins in PySpark and what are the different types of joins?
4. How do you perform data aggregation operations in PySpark?
5. What are window functions in PySpark and how are they used in DataFrames?
6. How do you select specific columns in a PySpark DataFrame?

7. How do you handle null or missing values in PySpark DataFrames?
8. How do you work with nested JSON structures in PySpark?
9. What is the difference between map() and flatMap() transformations in PySpark?
10. What are user-defined functions (UDFs) in PySpark and when do you use them?
11. How do you read data from CSV, JSON, and Parquet files in PySpark?
12. How do you integrate PySpark with libraries like Pandas and NumPy?
13. How do you work with custom data types in PySpark?
14. How do you use Spark SQL and execute SQL queries in PySpark?
15. What are the common data serialization formats and compression codecs used in PySpark?
16. How do you handle schema evolution in Spark when reading data with changing structures?
17. What are the various ways to select columns in a PySpark DataFrame?
18. How do you deal with missing or null values in PySpark DataFrames?
19. Provide examples of PySpark DataFrame operations that you frequently use.

5. Streaming & Real-Time Processing

1. What is structured streaming in PySpark and how does it work?
2. Explain the streaming capabilities of PySpark and how data is processed in real time.
3. How is fault tolerance ensured in Spark Streaming applications?
4. Describe the architecture of Spark Streaming and how it differs from Structured Streaming.
5. What are some real-time use cases where you used PySpark streaming?
6. Ensuring fault tolerance in a Spark Streaming application consuming data from Kafka - what strategies would you employ?
7. Can you elaborate on the use cases and benefits of using Apache Spark streaming for real-time data processing?

6. Machine Learning with PySpark

1. What is PySpark MLlib and what functionalities does it provide?
2. How do you use MLlib in PySpark for performing machine learning tasks?
3. How do you perform model evaluation and hyperparameter tuning in PySpark?
4. How do you implement large-scale machine learning using PySpark?
5. What are the key methods for optimizing ML pipelines in PySpark?
6. How do you handle iterative model training in distributed systems using PySpark?
7. How do you perform machine learning tasks using PySpark MLlib?

7. Deployment, Integration & Ecosystem

1. How do you deploy PySpark applications in a cluster environment?
2. How do you integrate PySpark with the Hadoop ecosystem (e.g., HDFS, YARN)?
3. What is the difference between managed and external tables in Spark?
4. Have you integrated PySpark with other technologies such as Kafka or NoSQL databases?
5. How do you transfer data between PySpark and external systems?
6. What tools or methods do you use for testing PySpark applications?
7. Have you integrated PySpark with other big data technologies or databases? If so, please provide examples.

8. Personal Experience & Behavioral

1. Can you provide an overview of your experience with PySpark?
2. What motivated you to specialize in PySpark?
3. Provide examples of PySpark operations that you have implemented in past projects.
4. How do you ensure data quality and consistency in your PySpark applications?
5. Which version control system do you use and how do you manage your PySpark codebase?